Experiment No : 6

Title: Write a program in C++ to perform following operations on complex numbers Add, Subtract, Multiply, Divide. Use operator overloading for these operations. The objective of this assignment is to learn the concepts of operator overloading.

Name of Student: Shamiksha Chaudhari
Class: SY **Div:** SY1 **PRN No:** B25ET1005 **Batch:** A1

Program:

```
#include<iostream>
#include<math.h>
using namespace std;
class Complex
{
private:
 double real, img;
public:
 void display();
 void read();
 Complex operator+(Complex);
 Complex operator-(Complex);
 Complex operator*(Complex);
 Complex operator/(Complex);
};

void Complex::read()
{
 cout << "Enter real part: ";
 cin >> real;
 cout << "Enter imaginary part: ";
 cin >> img;
}

void Complex::display()
{
 cout << real << " + " << img << " i " << endl;
}

Complex Complex::operator+(Complex C1)
{
 Complex Sum;
```

```
 Sum.real = real + C1.real;
 Sum.img = img + C1.img;
    return Sum;
}

Complex Complex::operator-(Complex C1)
{
 Complex Difference;
 Difference.real = real - C1.real;
 Difference.img = img - C1.img;
 return Difference;
}

Complex Complex::operator*(Complex C1)
{
    Complex Product;
    Product.real = (real * C1.real) - (img * C1.img);
    Product.img = (real * C1.img) + (img * C1.real);
    return Product;
}

Complex Complex::operator/(Complex C1)
{
    Complex Quotient;
    double denominator = (C1.real * C1.real) + (C1.img * C1.img);

    if (denominator == 0)
    {
        cout << "Error: Division by zero!" << endl;
        Quotient.real = 0;
        Quotient.img = 0;
        return Quotient;
    }
    Quotient.real = ((real * C1.real) + (img * C1.img)) / denominator;
    Quotient.img = ((img * C1.real) - (real * C1.img)) / denominator;
    return Quotient;
}



int main() {
    Complex C1, C2, C3, C4, C5, C6;

    cout << "Enter first complex number:" << endl;
```

```
    C1.read();

    cout << "Enter second complex number:" << endl;
    C2.read();

    cout << "Addition: ";
    C3 = C1 + C2;
    C3.display();

    cout << "Subtraction: ";
    C4 = C1 - C2;
    C4.display();

    cout << "Multiplication: ";
    C5 = C1 * C2;
    C5.display();

    cout << "Division: ";
    C6 = C1 / C2;
    C6.display();

    return 0;
}
```

Output :

Output

```
Enter first complex number:
Enter real part: 2
Enter imaginary part: 3
Enter second complex number:
Enter real part: 4
Enter imaginary part: 5
Addition: 6 + 8 i
Subtraction: -2 + -2 i
Multiplication: -7 + 22 i
Division: 0.560976 + 0.0487805 i
```